

# Seasonal Employment: Replication and Extension

Coglianesse & Price (2025) + QWI 6-Digit NAICS Index

May 13, 2026

# Outline

- 1 Replication of Coglianesse & Price (2025)
- 2 QWI 6-Digit NAICS Seasonal Index
- 3 Geographic Variation
- 4 Discussion

# Coglianese & Price (2025): Overview

- “Income in the Off-Season: Household Adaptation to Yearly Work Interruptions,” *ILRR* 2025
- **Question:** How do households smooth income when employment separations recur annually (i.e. are *seasonal*)?
- **Key innovation:** Identify seasonal separations from administrative data without requiring industry or occupation labels
- Core estimand: **excess recurrence** at 12-month horizon

$$\widehat{\text{ExR}} = \text{Pr}(\text{sep at } k=12) - \frac{\text{Pr}(\text{sep at } k=11) + \text{Pr}(\text{sep at } k=13)}{2}$$

- Counterfactual = average of adjacent lags  $\rightarrow$  isolates the *annual spike* marking genuine seasonal separations

## Event-study OLS on separators panel (CPS and SIPP):

$$\mathbf{1}[\text{sep at } k] = \sum_j \beta_j \mathbf{1}[k = j] + \varepsilon$$

- Estimate by OLS; excess recurrence =  $\beta_{12} - (\beta_{11} + \beta_{13})/2$
- Standard error via delta method
- **Two specifications:**
  - ① *Raw* (Figure 2 in paper): only horizon dummies, no constant
  - ② *Preferred* (Appendix F.1): adds controls for interview-seam artifacts (SIPP) and weeks elapsed since separation (CPS)
- Python replication uses `statsmodels` WLS with clustered SEs; data from the LEHD-style pre-cleaned `.dta.gz` files in the replication package

# Replication Results

Table: Excess recurrence estimates: paper vs. Python replication

| Specification       | Where    | Paper              | Replication |
|---------------------|----------|--------------------|-------------|
| CPS raw (Figure 2)  | Figure 2 | 1.4 p.p. (SE 0.14) | 1.44 ✓      |
| CPS preferred       | App. F.1 | 1.5 p.p. (SE 0.14) | 1.42 ✓      |
| SIPP raw (Figure 2) | Figure 2 | 2.0 p.p. (SE 0.11) | 2.04 ✓      |
| SIPP preferred      | App. F.1 | 1.6 p.p. (SE 0.11) | 1.62 ✓      |

- All four estimates within rounding of published values
- Industry ranking (both datasets): Agriculture > Entertainment > Education > Construction ≫ Healthcare, FIRE

# Extending to 6-Digit NAICS via QWI

- CP's classifier operates at the worker level; industry rankings are reported at 1-digit NAICS only
- **Goal:** produce a finer, industry-level seasonal index suitable for firm classification
- **Data:** Census Quarterly Workforce Indicators (QWI)
  - Quarterly separations and employment by state  $\times$  6-digit NAICS
  - 51 states + DC, years 2000–2023 (1,011 industries)
- **Quarterly analog to excess recurrence:** for each quarter  $q \in \{1, 2, 3, 4\}$  and industry  $j$ ,

$$\text{Excess}(q)_j = \text{SepRate}_{j,q,t} - \frac{\text{SepRate}_{j,q-1,t} + \text{SepRate}_{j,q+1,t}}{2}$$

(quarters treated cyclically; average over years  $t$ )

- **Peak excess:**  $\text{ExcessPeak}_j = \max_q \text{Excess}(q)_j$
- **Seasonal amplitude:**  $\max_q - \min_q$  normalised to  $[0, 1]$ ; does not assume any particular quarter is “the” seasonal quarter

# National Seasonal Index: Top Industries

**Table:** Top 12 most seasonal 6-digit NAICS industries (national)

| NAICS & description           | ExcessPeak | $n$ |
|-------------------------------|------------|-----|
| 111160 Leafy veg. farming     | 0.561      | 22  |
| 111219 Other berries          | 0.454      | 14  |
| 115113 Crop harvesting        | 0.398      | 10  |
| 111199 Other food crops       | 0.397      | 23  |
| 111140 Wheat farming          | 0.287      | 20  |
| 111336 Fruit & tree nut farm. | 0.333      | 18  |
| 115111 Cotton ginning         | 0.414      | 2   |
| 311421 Fruit/veg. canning     | 0.466      | 21  |
| 711212 Racetracks             | 0.371      | 21  |
| 115114 Postharvest services   | 0.382      | 22  |
| 111998 Other crop farming     | 0.328      | 20  |
| 111120 Oilseed farming        | 0.345      | 12  |

- $n$  = years contributing to ExcessPeak estimate; small  $n$  reflects Census disclosure suppression for thin cells
- Non-agriculture entries: fruit/veg canning (311421), racetracks (711212) — both with  $n = 21$ , robust estimates
- Seasonal index normalized to  $[0,1]$  via 99th-pct winsorization
- Correlation with CP 1-digit rankings:  $\rho = 0.826$

# National Seasonal Index: Sector Summary

Table: Mean seasonal index by 2-digit sector (national)

| Sector                           | Mean seasonal index | ExcessPeak (p.p.) |
|----------------------------------|---------------------|-------------------|
| Agriculture, forestry, fishing   | 0.472               | 15.2              |
| Arts, entertainment & recreation | 0.419               | 12.1              |
| Educational services             | 0.257               | 8.4               |
| Construction                     | 0.183               | 5.9               |
| Personal services                | 0.126               | 4.2               |
| Business services (54–56)        | 0.123               | 4.2               |
| Finance & real estate            | 0.121               | 4.2               |
| Retail trade                     | 0.115               | 3.5               |
| Health care & social assistance  | 0.063               | 2.2               |
| Wholesale trade                  | 0.060               | 1.8               |
| Manufacturing (metals/machinery) | 0.034               | 1.1               |

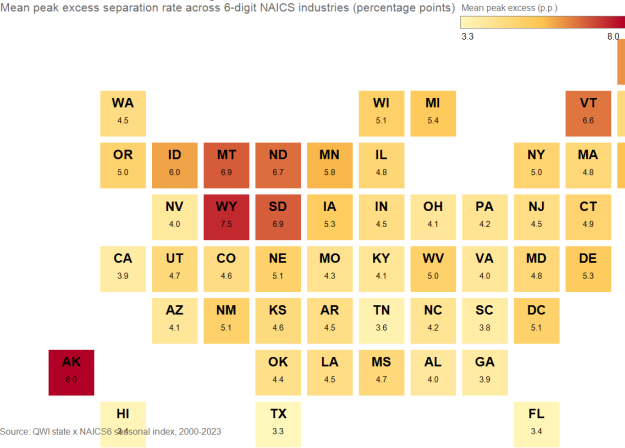
Education peaks in Q2 (end of school year), not Q4 — ExcessPeak is positive across all sectors. CP's 1-digit rankings confirmed at finer granularity.

# Geographic Variation: Overview

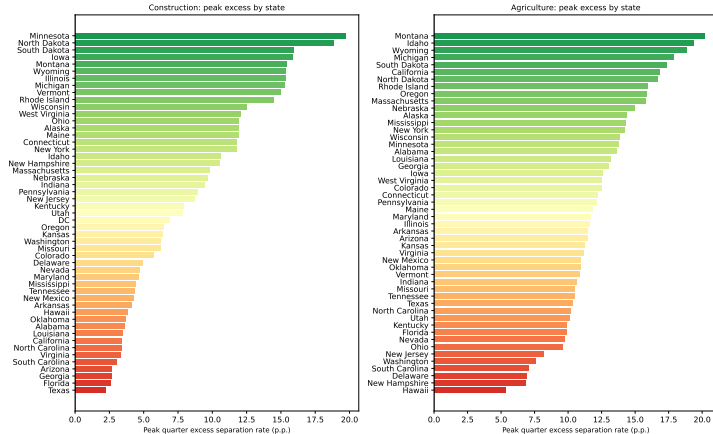
- The national index masks substantial state-level heterogeneity
- We compute ExcessPeak for each state  $\times$  6-digit NAICS cell (40,222 cells with  $\geq 5$  year-observations)
- **Main finding:** a  $2.4\times$  gap between most and least seasonal states in mean peak excess (Alaska: 8.0 p.p. vs. Texas: 3.3 p.p.)
- Northern/Great Plains states consistently more seasonal across multiple sectors; Sun Belt states consistently less so
- **Most geographically variable sectors:**
  - 1 Agriculture (cross-state SD = 15.4 p.p.)
  - 2 Arts & entertainment (SD = 14.4 p.p.)
  - 3 Construction (SD = 9.9 p.p.)
- **Least geographically variable:** healthcare, metals manufacturing, utilities — consistent low seasonality everywhere

# State-Level Mean Seasonality

## State-level seasonality



# Construction and Agriculture by State



**Construction:** Minnesota (19.7 p.p.) vs. Florida (2.6 p.p.) — an 8× gap driven by winter construction shutdowns in northern states. **Agriculture:** Montana (20.2 p.p.) and Idaho (19.3 p.p.) lead.

# State-Specific Top and Bottom 1% Industries

**Table:** Within-state NAICS6 tails

| Statistic               | Top 1% | Bottom 1% |
|-------------------------|--------|-----------|
| State-industry slots    | 413    | 413       |
| Unique NAICS6 codes     | 133    | 252       |
| Codes appearing once    | 57     | 158       |
| Codes in 10+ states     | 6      | 0         |
| Pairwise Jaccard        | 0.054  | 0.015     |
| Also national same tail | 12.6%  | 2.7%      |

Interpretation: the state-specific tails are related to the national index, but most state-tail industries are not simply the national top or bottom 1% repeated everywhere.

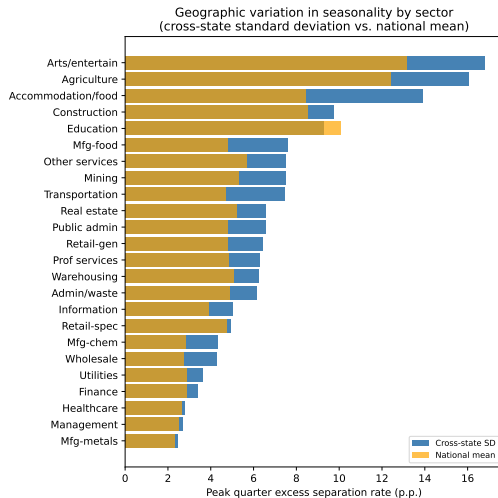
## Recurring top-1% industries

- Nursery/tree production: 15 states
- Marinas: 14 states
- Specialty trade contractors: 12 states
- Golf courses/country clubs: 11 states
- Miscellaneous crop farming: 11 states

## Bottom-1% pattern

- More diffuse than the top tail
- Concentrated in metals manufacturing, wholesale, chemicals, finance, and transportation

# Geographic Variation: Sector Cross-State SD



# Implications for Firm Classification

- The QWI index enables firm-level seasonal classification via NAICS lookup
- The top/bottom 1% analysis shows the exact NAICS6 tails differ sharply across states: top tail has 133 unique codes; bottom tail has 252
- **But:** for geographically sensitive sectors (construction, agriculture), a single national index may misclassify:
  - A Minnesota construction firm is highly seasonal
  - A Florida construction firm is nearly non-seasonal
- **Preferred approach for firm classification:**
  - ① Use the  $state \times NAICS$  ExcessPeak as the lookup value when state of operation is known
  - ② Fall back to national index for industries with thin state-level coverage (low  $n_{excess\_obs}$ )
  - ③ Direct estimation from UI wage records when sufficient separations are available ( $n \geq 20$  per firm)
- **Caveat:** 6-digit cells are often suppressed in small states; state-level estimates for niche agriculture sub-industries may rest on very few year-observations

# Summary

- 1 **Replication:** Python replication of CP (2025) excess recurrence matches all four published estimates within rounding
- 2 **National index:** QWI-based ExcessPeak at 6-digit NAICS correlates strongly with CP's 1-digit rankings ( $\rho = 0.826$ ); agriculture and entertainment top the seasonal hierarchy
- 3 **Geographic variation:** large and systematic state heterogeneity — a  $2.4\times$  gap between most and least seasonal states (Alaska vs. Texas); state-specific top/bottom 1% NAICS lists differ sharply from each other and from the national tails
- 4 **Next step:** apply state  $\times$  NAICS index to UI wage records for firm-level seasonal classification